

Neural Network Architecture and Performance Analysis

1. Architecture Design Experiment (Task 2)

To understand how structural changes impact learning, three Multilayer Perceptron (MLP) architectures were evaluated on the FashionMNIST dataset.

- **Performance Comparison:** Among the unnormalized models, **Model B** (\$784 \rightarrow 128 \rightarrow 64 \rightarrow 10\$) performed best, achieving a test accuracy of **87.41%**. It successfully balanced learning capacity with generalization.
- **The Effect of Depth:** Increasing network depth does not guarantee better performance. Model C was deeper than Model B but resulted in a slightly lower accuracy (87.11%). In relatively simple datasets, excessive depth can lead to overfitting or optimization challenges like vanishing gradients, making deeper models underperform compared to well-tuned shallower ones.
- **The Effect of Parameter Count:** Parameter count directly controls a model's learning capacity. Model A (25,450 parameters) lacked the capacity to capture all data patterns, yielding the lowest accuracy (86.60%). Model B (109,386 parameters) provided the optimal capacity. Model C (243,082 parameters) had too much capacity, allowing it to memorize noise in the training set, which slightly degraded its test accuracy.

2. Underfitting vs. Overfitting Analysis (Task 3)

Two extreme architectures were tested to observe learning failures.

- **Underfitting (Small Model):** The model with only one hidden layer of 4 neurons (\$784 \rightarrow 4 \rightarrow 10\$) severely underfit the data. It capped out at a Train Accuracy of **62.77%** and a Test Accuracy of **62.10%**. Looking at the loss curves, both training and validation loss plateaued early at very high values. The model simply lacked the structural complexity to map the inputs to the correct outputs.
- **Overfitting (Large Model):** The highly complex model (\$784 \rightarrow 1024 \dots \rightarrow 10\$) exhibited severe overfitting. It achieved a near-perfect Train Accuracy of **94.97%**, but its Test Accuracy stalled at **88.72%**. The loss curves perfectly illustrate this: the training loss quickly plummeted toward zero, while the validation loss initially

dropped but then stalled and began fluctuating. This widening gap confirms the model memorized the training set but failed to generalize to unseen data.

3. Batch Normalization Study (Task 4)

Batch Normalization (BN) was added to Model B to evaluate its impact on training dynamics.

- **Convergence and Stability:** Adding BN significantly accelerated convergence, allowing the model to reach lower loss values in fewer epochs. It also stabilized the training process by smoothing out fluctuations in the loss curve, ultimately pushing the model to the highest overall test accuracy of **88.45%**.
- **Parameter Count:** BN normalizes layer inputs and introduces two learnable parameters (γ for scaling and β for shifting) per node. This resulted in a slight parameter increase:
 - *Without BN:* 109,386 parameters
 - *With BN:* 109,770 parameters

4. Error Analysis (Task 5)

An analysis of the confusion matrix reveals the inherent limitations of the MLP architecture on image data.

- **Frequently Misclassified Labels:** The model struggled most with visually similar categories. Shirt, T-shirt/top, and Pullover were frequently confused with one another. Another major overlap occurred between Sneaker and Ankle boot.
- **Cause of Errors:** These garments share nearly identical silhouettes, edges, and pixel intensities in a 28x28 grayscale format. Furthermore, MLPs process images by flattening them into a 1D vector. This destroys the 2D spatial hierarchy and local pixel relationships (e.g., the proximity of a collar to a sleeve), making it exceedingly difficult for the network to distinguish fine structural details.

5. Model Efficiency Comparison (Task 6)

Model	Parameters	Accuracy	Training Time (s)

Model A	25,450	86.60%	185.05
Model B	109,386	87.41%	189.04
Model C	243,082	87.11%	212.92
Model B (BN)	109,770	88.45%	196.88

Data extracted from notebook execution.

The Complexity Trade-off: The table highlights the core trade-off in deep learning. Upgrading from Model A to Model B required 4x more parameters but yielded a solid accuracy boost with minimal time penalty. However, pushing complexity further to Model C more than doubled the parameters again and increased training time to 212 seconds, yet accuracy *decreased* due to overfitting. The most efficient solution was utilizing a medium-complexity model (Model B) paired with Batch Normalization, which delivered the highest accuracy for a moderate computational cost.

6. Gradient Stability Investigation (Bonus Task)

An experiment utilizing an 8-hidden-layer MLP was conducted to observe how extreme depth affects backpropagation.

- **Results:**
 - 8-Layer MLP **without** BN: **86.53%** Accuracy
 - 8-Layer MLP **with** BN: **87.52%** Accuracy
- **The Vanishing Gradient Problem:** In deep networks without normalization, error signals are repeatedly multiplied by small weights during backpropagation. This causes the gradients to shrink exponentially, meaning the earliest layers receive almost zero

updates and fail to learn. This poor "gradient flow" stalled the unnormalized network's training.

- **Mitigation via Batch Normalization:** BN solves this by recentering and rescaling the activations at every hidden layer. By ensuring inputs to the activation functions remain in the "active" non-saturated regions (e.g., the linear region of a ReLU or Sigmoid), BN preserves the magnitude of the error signal. This ensures a healthy, consistent gradient flow across all 8 layers, allowing the network to train effectively and boosting accuracy by roughly 1%.